Methods of proof – 2

Induction: strengthening, dimension, descent

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Theoretical minimum.

Fact 1 (strengthen the statement). A statement is often easier to prove in a stronger form, because the induction hypothesis becomes stronger too. The standard example is an inequality with a constant: replace the constant by an explicit expression in n and the induction closes. If the step does not close, the correct reaction is to sharpen the statement, not to look for a longer computation.

Fact 2 (what to induct on). Besides the number of objects: the dimension of a space, the rank of a matrix, the degree of a polynomial, the number of prime factors, the size of a Jordan block, the number of vertices of a graph, the number of "bad" elements. Two parameters give a double induction, and a lexicographic order on pairs makes it a single one.

Theorem 1 (descent and the minimal counterexample). Every non-empty set of positive integers has a least element; hence a counterexample of least size exists and may be assumed to be minimal, and a construction producing a strictly smaller counterexample proves the statement. Over the reals this fails, and its place is taken by compactness (Methods of proof – 3).

Theorem 2 (transfinite induction and Zorn's lemma). Every partially ordered set in which every chain has an upper bound has a maximal element. This is what produces a Hamel basis (Continuous functions – 6), a basis of an arbitrary vector space, a maximal ideal, and an extension of a linear functional; it is not needed when the object can be built by a countable procedure.

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1. Prove by induction with a strengthened statement: $\sum_{k=1}^{n} \frac{1}{k^2} \leqslant 2 - \frac{1}{n}$, and then $\prod_{k=1}^{n} \frac{2k-1}{2k} \leqslant \frac{1}{\sqrt{3n+1}}$; in both cases explain why the naive statement does not close and the sharpened one does.

2. Prove by induction on the dimension that every symmetric bilinear form over a field of characteristic different from 2 has an orthogonal basis, and by induction on the rank that a matrix of rank r is a sum of r matrices of rank one (compare with Linear algebra – 2).

3. Prove Helly's theorem in $\mathbb{R}^n$ by induction on the number of sets, and explain which step uses the dimension and which uses the number of sets (compare with Convex – 1).

4. Prove by infinite descent that $\sqrt{2}$ is irrational, that $x^4 + y^4 = z^2$ has no positive integer solutions, and that a minimal counterexample argument proves the fundamental theorem of arithmetic.

5. Prove Zorn's lemma implies the existence of a basis in every vector space, and use transfinite induction to construct a set $A \subseteq \mathbb{R}^2$ meeting every line in exactly two points.